

Exercise 5: Recap of Elementary Probability and Statistics

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Problem 1 (Mean Squared Error (MSE) and Estimator Comparison):

Let $X_1, X_2, \dots, X_n$, $n \geq 2$, be a random sample from a distribution with mean μ and variance $\sigma^2 = 1$. Consider two estimators for the population mean μ :

1. $\hat{\mu}_1 = \bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ (the sample mean)
2. $\hat{\mu}_2 = \frac{X_1 + X_n}{2}$ (the average of the first and last observations)

- (a) Calculate the bias of both estimators. Are they unbiased?
- (b) Calculate the variance of both estimators.
- (c) Calculate the Mean Squared Error (MSE) for both estimators.
- (d) Based on their MSE, which estimator is better? Is there any scenario (i.e., a specific value of n) where you might prefer $\hat{\mu}_2$? Explain your reasoning.

Problem 2 (The Cramér-Rao Lower Bound (CRLB) and Efficiency):

Let $X_1, X_2, \dots, X_n$ be an i.i.d. sample from a Poisson distribution with parameter λ . The probability mass function (PMF) is given by $\mathbb{P}(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$ for $k = 0, 1, 2, \dots$. The standard estimator for λ is $\hat{\lambda} = \bar{X}_n$, where $\bar{X}_n$ is the sample mean.

- (a) Calculate the Fisher Information, $I(\lambda)$, from a single observation.
- (b) Determine the **Cramér-Rao Lower Bound** for the variance of any unbiased estimator of λ .
- (c) Calculate the variance of the estimator $\hat{\lambda} = \bar{X}_n$.
- (d) Compare the variance of $\hat{\lambda}$ to the **CRLB**. Is $\bar{X}_n$ an efficient estimator for λ ?

Problem 3 (MSE in a Mixture Model):

A company produces microchips at two factories, A and B. The time-to-failure (in years) for a chip from Factory A follows an Exponential distribution with mean $\theta_A = 5$. The time-to-failure for a chip from Factory B follows an Exponential distribution with mean $\theta_B = 2$. A shipment contains chips where 25% are from Factory A and 75% are from Factory B. Let X be the lifetime of a single randomly drawn chip.

- (a) What is the true expected lifetime, $\mu = \mathbb{E}[X]$?
- (b) Prove the law of total variance: if X and Y are random variables on the same probability space, and the variance of Y is finite, then:

$$\text{Var}(Y) = \mathbb{E}[\text{Var}(Y|X)] + \text{Var}(\mathbb{E}[Y|X]),$$

where $\text{Var}(Y|X) := \mathbb{E}[Y^2|X] - (\mathbb{E}[Y|X])^2$.

- (c) Find the population variance, $\sigma^2 = \text{Var}(X)$.
- (d) Let $\hat{\mu} = \bar{X}_n$ be the sample mean from a sample of n chips. What is the MSE of this estimator for μ ?

Problem 4 (Bayesian Estimation vs. Frequentist Estimation):

You are presented with a coin that is either fair ($p = 0.5$) or biased towards heads ($p = 0.7$). You assign a prior probability of 50% to each case. Let θ be the true probability of heads, so $\mathbb{P}(\theta = 0.5) = 0.5$ and $\mathbb{P}(\theta = 0.7) = 0.5$. You flip the coin $n = 10$ times and observe $k = 8$ heads.

- (a) What is the Maximum Likelihood Estimate (MLE) for θ , denoted $\hat{\theta}_{MLE}$?
- (b) Use Bayes' Theorem to calculate the posterior probabilities, $\mathbb{P}(\theta = 0.5|\text{data})$ and $\mathbb{P}(\theta = 0.7|\text{data})$.
- (c) Calculate the Bayes estimator for θ using the posterior mean, $\hat{\theta}_{Bayes} = \mathbb{E}[\theta|\text{data}]$.
- (d) Compare your two estimates. What would happen to the Bayes estimator if you had observed $n = 1000$ flips with $k = 800$ heads? Explain.

Problem 5 (Estimator for a Uniform Distribution):

Let $X_1, X_2, \dots, X_n$ be an i.i.d. random sample from a **Uniform distribution** on the interval $[0, \theta]$, where $\theta > 0$. The mean of this distribution is $\mathbb{E}[X_i] = \frac{\theta}{2}$ and the variance is $\text{Var}(X_i) = \frac{\theta^2}{12}$.

A common estimator for the parameter θ is the Method of Moments Estimator, which is $\hat{\theta} = 2\bar{X}$, where $\bar{X}$ is the sample mean.

- (a) Show that $\hat{\theta}$ is an unbiased estimator for θ .
- (b) Find the variance of the estimator, $\text{Var}(\hat{\theta})$.
- (c) Suppose the true parameter is $\theta = 10$ and our sample size is $n = 48$. Use the CLT to approximate the probability that our estimate will be accurate within 0.5 of the true value (use that $\Phi(0.6) \approx 0.7257$).